



Computer Science Department
COMP233-Discrete Mathematics/ Spring 2026
Section #4 - Quiz #1 (14 April): Chapters 1 & 2

Student Name:

Student ID:

Question #1: [45 Marks]

- a) Simplify the following logical expression using the laws of logic. You must state the name of the law used in each step: [15 pts]

$$\neg(p \rightarrow q) \vee (p \wedge r)$$

- b) Show that the following expression is a Tautology by proving the logical equivalence using a truth table: [25 pts]

$$((p \rightarrow q) \wedge \neg q) \rightarrow \neg p \equiv T$$

- c) Which Rule of Inference is represented by the tautology in **part (b)**? Write its formal argument form. [5 pts]

Question #2: [55 Marks]

- a) **Consider the following statement regarding the Birzeit University academic system:** [35 pts]

"Entering the correct password on Ritaj is a **necessary condition** for every student to check their Discrete Math marks or register for the course."

Use the following predicates and domain:

- **P(x):** "x enters the correct password on Ritaj"
- **M(x):** "x checks Discrete Math marks"
- **R(x):** "x registers for the course"
- **Domain of discourse:** All students.

Task: Based on the statement above, complete the table in the **answer sheet** by providing the Logical Form (using formal symbols and quantifiers), and the corresponding verbal sentence only for each Contrapositive and Negation of the logical variations.

- b) **Provide the direct logical equivalent or operation for each of the following (Write the formal expression):** [20 pts]

- 1) Write the symbolic negation of: $\forall x \exists y, P(x, y)$, What is the Arity of the predicate P?
- 2) Given $p \rightarrow q$, what is the contrapositive of its converse?
- 3) Write the formal argument form of the Hypothetical Syllogism (Transitivity).

Please, use the answer sheet form ONLY!

Good Luck!



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Answer Sheet Question #1

Question #1:

[45 Marks]

a) Expression: $\neg (p \rightarrow q) \vee (p \wedge r)$

[_____/15 pts]

Step	Expression	Law Used

b) Expression: $((p \rightarrow q) \wedge \neg q) \rightarrow \neg p \equiv T$

[_____/25 pts]

p	q					

Conclusion:

c)

[_____/5 pts]

Name of the Inference Rule _____

Formal Argument Form:

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. .
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Answer Sheet Question #2

Question #2:

[55 Marks]

a)

[____/35 pts]

Concept	Logical Form	Verbal Statement
1. Formalization:		Entering the <u>correct password</u> on Ritaj is a necessary condition for <u>every student</u> to check their <u>Discrete Math marks</u> or <u>register for the course</u> .
2. Converse:		XXXXXXXXXX
3. Inverse:		XXXXXXXXXX
4. Contrapositive:		
5. Negation:		

b)

[____/20 pts]

1. Answer: The symbolic negation is: _____

Arity #:

2. Answer: The contrapositive of its converse is:

3. Answer: The formal argument of the Hypothetical Syllogism (Transitivity) is:

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Answer Sheet